

AWS ECR (Elastic Container Registry)

01-ml-infrastructure/ecr

Overview

ECR is AWS managed container image storage. It holds versioned Docker and OCI images that other services can pull when they need to start a container.

For ML teams, ECR is where custom training images, inference images, and utility containers are stored so that deployments are reproducible and centrally managed.

Core Concepts

- Repositories group related images, while tags and digests identify specific builds. Digests are immutable and better for strict reproducibility than floating tags.
- ECR supports vulnerability scanning, lifecycle policies, encryption, and repository policies for cross account sharing.
- Authentication is integrated with IAM. Build systems and runtimes usually obtain temporary credentials and then push or pull images through the ECR API.

How It Works

- A CI system builds the image, runs tests, tags it, and pushes it to an ECR repository.
- Services such as ECS, SageMaker, or Lambda container images then pull the image when creating or updating a workload.
- Multi account setups often replicate approved images from a build account into runtime accounts to separate software production from software consumption.

AI and ML Relevance

- ML inference containers often include model servers, preprocessing logic, and optimized native libraries. ECR ensures those dependencies are packaged consistently across environments.
- Training images in ECR let you standardize frameworks, CUDA libraries, and internal tooling for repeatable experiments and pipelines.

Operational Notes

- Avoid using mutable tags such as latest as the only deployment reference. Use digests or versioned tags in release automation.
- Enable lifecycle policies so old image layers do not accumulate forever and quietly increase storage cost.
- Scan images, patch base layers regularly, and keep the build chain deterministic so security and debugging stay tractable.

What To Remember

- ECR stores images; ECS, EKS, SageMaker, and Lambda run them.

- Tagging is for convenience. Digests are for certainty.
- Treat the container registry as part of the software supply chain, not just as storage.